Mapping Vulnerability:
House Income and Seismic Hazard Exposure Across the San Francisco Bay Area

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CYPLAN 101: Introduction to Urban Data Analytics

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1. Research Question

Restate your research question (including the geography, population, and time frame it covers) and explain what has changed since your proposal and why. Changes might include a narrower scope, a different dataset, or a reformulated question.

Across the nine-county San Francisco Bay Area, do Census tracts with lower household incomes and higher concentrations of affordable housing have a greater percentage of their land area within mapped seismic hazard zones?

Since our proposal, we have narrowed our analysis to the nine-county Bay Area and shifted from a broader county-level analysis to a Census tract-level analysis. We plan to map CTCAC affordable housing projects to Census tracts and combine them with household income data at a finer geographic scale. We also updated our income data from the 2014–2018 CHAS dataset to the 2018–2022 dataset and are considering ACS C17002 for more detailed poverty measures. These changes allow us to better examine whether lower-income communities and areas with more affordable housing are disproportionately located within seismic hazard zones.

2. Data Cleaning Process

Summarize your data cleaning process and describe what you removed, filtered, or fixed, and how many records were affected. For example: "We dropped 1,204 parcels with missing geometry, about 3% of rows." There should also be a justification for every cleaning step.

For our project, we are using data from the California Tax Credit Allocation Committee (CTCAC) and HUD Comprehensive Housing Affordability Strategy (CHAS) to study affordable housing and household income across the nine Bay Area counties.

We started by cleaning the CTCAC dataset, which contains information about affordable housing projects. We filtered the original statewide data to the nine Bay Area counties, leaving 1,510 project records. We also made the location names more consistent and converted the low-income housing unit counts into numerical values. During cleaning, we found 72 records (4.8%) with missing unit counts. We kept these records with their missing values because replacing them with zero could affect the results. We also corrected the name "Pt. Reyes Station" to "Point Reyes Station," which allowed two additional projects to match the Census boundaries. One project remained unmatched because its location was only listed as Marin County.

For household income, we first worked with the 2014–2018 CHAS dataset and removed 11 rows containing missing values. We later used the updated 2018–2022 CHAS dataset, which originally contained 3,222 records. We checked the data types to make sure the numerical columns were correctly formatted and found no missing values in the loaded Table 1 data. We then filtered the dataset to the nine Bay Area counties and created a stacked bar chart showing the share of households in different income categories. We also checked the calculated household totals

against the original values. The differences were small, with a maximum of five households per county.

Our group is also considering American Community Survey (ACS) data, particularly table C17002, to study poverty levels at a smaller geographic scale. This would allow us to compare household economic conditions with affordable housing locations and mapped earthquake hazard areas.

3. Methodology & Techniques

Walk us through your current workflow step by step, naming the core operations you rely on (e.g., spatial joins, buffers, aggregations) and the projected CRS you use, with its EPSG code.

Our analytical workflow integrates spatial data processing, geometric transformations, and spatial joins to evaluate the proximity and overlap of subsidized affordable housing relative to seismic hazard zones and historic earthquake epicenters across the nine Bay Area counties.

To ensure geometric accuracy in spatial measurements, all vector datasets were reprojected from their native geographic coordinate systems (EPSG:4326) into a unified projected coordinate system: California Albers Equal Area (EPSG:3310). This projection minimizes area distortion across California, making it optimal for polygon area calculations and spatial joins.

Datasets were filtered spatially using county boundary polygons to strictly isolate the nine San Francisco Bay Area counties. Invalid geometries were repaired using shapely.validation.make_valid, and missing or inconsistent text fields were standardized. We performed point-in-polygon and proximity spatial joins to connect distinct data layers. Specifically, affordable housing project point locations were spatially joined with Alquist-Priolo earthquake fault hazard zones and buffered historic earthquake occurrence zones to determine whether affordable housing units disproportionately intersect high-risk seismic areas.

Neighborhood and county-level boundaries were spatially aggregated to calculate metrics such as the total number of subsidized low-income units per geographic unit and the percentage of neighborhood land area situated within active seismic hazard zones.

4. Early Results

What patterns do you notice in your analysis so far? Are you surprised by what you've found? Why or why not?

Our preliminary results show that affordable housing exposure to historic earthquakes depends heavily on the distance threshold. Of 1,472 mapped CTCAC projects, none are within 5 km of a historic M5+ epicenter, 38 are within 10 km, and 490 are within 25 km. Exposure is concentrated in the South Bay, with Santa Clara accounting for 25 of the 38 projects within 10 km.

Income patterns are less distinct at the county level: the share of households below 80% HAMFI ranges from 38-45%. However, renters are consistently more likely to be low-income than homeowners across all counties. We were surprised that San Francisco had no projects within 10 km of a historic epicenter despite its proximity to the San Andreas fault. This reinforces that our limited historical earthquake catalog is better used as a validation measure than as our primary measure of hazard.

5. Plan to Finish

Lay out the work remaining between now and the final deadline, with who is responsible for each step and a target date. Identify the biggest risks to finishing, such as missing data, slow processing, or a method that isn't working, and address how you can adapt your project as needed. Close with 1–3 specific challenges your team is facing or aspects of your project you would like feedback on.

Nov. 3–22: Complete the tract-level spatial analysis, integrate hazard exposure with socioeconomic and affordable-housing data, and finalize visualizations and robustness checks. Erin and Aalini will focus on the spatial and socioeconomic analysis, Anastasiia on affordable housing, and Katie on earthquake validation.

Nov. 23–Dec. 5: Build the GitHub Pages website, led by Katie, while each team member contributes their analysis, visualizations, and written interpretation.

Dec. 6–15: Review the full project, verify results, refine the website, and finalize conclusions and limitations.

Our main risk is integrating datasets that use different geographic scales and identifiers. If tract-level variables cannot be reliably merged, we will prioritize measures with consistent geographic coverage. We would especially appreciate feedback on whether our exposure measure is appropriate and how best to combine tract-level socioeconomic data with point-based affordable-housing data.